Signals and Systems time domain representation and Signal Operations

Participated in & did the coursework [Y/N]?

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1 Introduction

Representing signals in the time domain is essential for analyzing how systems respond to external inputs in continuous-time signal processing [1]. Aligned with that, the primary objective of this laboratory exercise is to mathematically define, manipulate, and visualize continuous-time signals and systems. Where the experiment aims to apply fundamental time-domain operations including time shifting, time flipping, and time scaling to given piecewise sequences. The exercise also aims to determine and plot the impulse responses of ideal frequency-selective filters (lowpass, highpass, bandpass, and bandstop) and evaluate the auto-correlation of a given signal to identify its temporal characteristics.

These objectives are grounded in the fundamental theory of linear time-invariant (LTI) systems, which can be analyzed through three interrelated concepts: temporal manipulation, system characterization, and internal signal analysis. First, regarding temporal manipulation, time-domain operations fundamentally alter a signal's independent variable [1]. Specifically, time shifting advances or delays the waveform along the time axis, time flipping reflects it across the origin, and time scaling compresses or expands its duration [2]. Second, regarding system characterization, an LTI system's behavior is entirely defined by its impulse response, which dictates the system's reaction to an instantaneous unit impulse [3]. Evaluating these responses for specific filters yields critical insight into their transient behavior and frequency selectivity [3]. Third, regarding internal signal analysis, comprehensive evaluation necessitates examining a signal's inherent properties independent of external filters [4]. Rather than observing how a signal interacts with a system, analysts must observe how a signal interacts with itself over time. Auto-correlation provides the exact mathematical mechanism for this internal comparison, measuring the similarity between a waveform and its own time-delayed counterpart to detect hidden periodicities or structural lags [4].

To computationally achieve these theoretical objectives, a specific, organized suite of MATLAB functions is required. For precise mathematical modeling, the syms function is utilized to declare symbolic variables. For signal construction, piecewise continuous sequences and system excitations are generated using the heaviside (unit step) and dirac (unit impulse) commands, which are subsequently augmented by exp and cos to model exponential decay and sinusoidal oscillations. For statistical computation, the xcorr function is used to calculate the discrete auto-correlation sequence. Finally, for data visualization, figure, subplot, and plot are employed to generate multi-axis graphs that seamlessly compare the manipulated signal sequences and filter responses.

2 Experimental Design

The experimental design is structured to systematically investigate the properties of continuous-time signals through mathematical modeling and computational simulation. By leveraging MATLAB's Symbolic Math Toolbox, this methodology demonstrates how theoretical signal operations can be implemented and validated within a software environment. [cite: 5, 277, 511]

2.1 Signal Construction and Transformation Methodology

The core of this experiment involves the symbolic definition of a baseline signal, $x(t)$, which serves as the primary subject for various time-domain operations. [cite: 9, 281, 515] The signal is constructed using Heaviside step functions to isolate its behavior within the specific interval of $t = 0$ to $t = 8$ seconds. The exact symbolic math expression designed for this experiment is: [cite: 29, 298, 535]

$$x(t) = (1 - 0.25t) \cdot (\text{heaviside}(t) - \text{heaviside}(t - 8)) \quad (1)$$

The experimental procedure for signal transformation required the modification of the independent variable t to achieve specific outcomes: [cite: 32, 300, 541]

- **Time Shifting**: Replacing t with $(t-2)$ for a delay and $(t+4)$ for an advance. [cite: 39, 45, 308, 314, 553, 560]
- **Time Reversal**: Substituting t with $(-t)$. [cite: 50, 321, 569]

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- **Combined Operations**: Implementing $x(3-t)$ to achieve both flipping and shifting. [cite: 54, 327, 577]
- **Time Scaling**: Multiplying t by a constant factor of 2 ($x(2t)$) to achieve compression. [cite: 60, 333, 584]

These transformations were consolidated into a single figure using a 3×2 subplot configuration to allow for simultaneous comparative analysis of the six resulting waveforms. [cite: 32, 300, 542]

2.2 Impulse Response Modeling for Filter Systems

The methodology also addressed the representation of system characteristics through the definition of impulse response functions ($h(t)$). [cite: 141, 417, 647] Four distinct filter types were modeled using combinations of the Dirac delta function ($\delta(t)$) and the Heaviside step function ($u(t)$) to simulate ideal signal processing behavior. [cite: 147, 157, 424, 430, 656, 664] The specific equations implemented for these models are summarized in Table 1. [cite: 143, 149, 153, 157, 421, 424, 427, 430, 650, 656, 660, 664]

Table 1
Mathematical Expressions for Filter Impulse Responses

Filter Type	Symbolic Equation	MATLAB Implementation
Lowpass	$h_1(t) = e^{-10t}u(t)$	`exp(-10*t) * heaviside(t)`
Highpass	$h_2(t) = \delta(t) - e^{-10t}u(t)$	`dirac(t) - exp(-10*t) * heaviside(t)`
Bandpass	$h_3(t) = e^{-10t}\cos(500t)u(t)$	`exp(-10*t)*cos(500*t)*heaviside(t)`
Bandstop	$h_4(t) = \delta(t) - e^{-10t}\cos(500t)u(t)$	`dirac(t) - exp(-10*t)*cos(500*t)*heaviside(t)`

2.3 Numerical Computation and Statistical Correlation

To determine the auto-correlation of the signal $x(t)$, the methodology required a transition from symbolic expressions to numerical data arrays. [cite: 193, 469, 694] This was achieved by defining a discrete time vector ranging from -10 to 10 seconds with a high-resolution step size of $dt = 0.01$. [cite: 194, 197, 470, 471, 696, 697] The symbolic expression $x(t)$ was evaluated over this vector to create the numerical array `x_num`. [cite: 202, 472, 698] The auto-correlation was computed using the `xcorr` function, and the resulting sequence was normalized by multiplying by the time step dt. [cite: 218, 233, 473, 474, 699, 700]

2.4 Experimental Pseudocode

The following logic describes the programmatic implementation of the experiment: [cite: 32, 160, 193, 300, 432, 469, 542, 667, 694]

```
1. START
2. DEFINE symbolic variable 't'
3. CONSTRUCT signal x(t) using Heaviside functions for interval [0, 8]
4. FOR EACH transformation (shift, flip, scale):
     APPLY transformation to t
     PLOT result in 3x2 subplot grid (Figure 1)
5. CONSTRUCT signal y(t) = exp(-t)u(t) + u(-1-t)
6. PLOT y(t) for t in range [-6, 6] (Figure 2)
7. DEFINE h1(t) through h4(t) for LPF, HPF, BPF, and BSF
8. OVERLAY h1(t) and h2(t) (excluding delta) on same axis (Figure 3)
9. CONVERT symbolic x(t) to numeric array x_num with dt = 0.01
10. COMPUTE Rxx = xcorr(x_num, x_num) * dt
11. PLOT Rxx vs tau (Figure 4)
12. END
```

3 Analysis of Results

Able to consolidate the results of each group member.

4 Discussion and Conclusion

Able to discuss the overall story of the experiment. The discussion and conclusion are linked to the objectives, experiment, and results

5 Authors Contribution

On what contributions to specify, see the terms at www.elsevier.com.

1) BAUTISTA, Alyssa Nicole : Methodology, Software, Validation, Formal Analysis, Writing – Original Draft, and Review and Editing, Visualization, Project Administration
2) GONZALES, Jeremia E.: Conceptualization, Software, Investigation, Resources, Data Curation, Writing – Original Draft, and Review and Editing, Supervision
3) MENDIOLA, Keifer Judd: Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, and Review and Editing

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